

What Is a Force?

A force is a push or a pull — and it can start, stop, speed up, slow down, turn, or reshape an object.

Force: a push or a pull

A **force** is a **push or a pull** on an object that results from its **interaction** with another object. At least **two objects** must interact for a force to act.

SI unit: the unit of force is the **newton** (small "n"), symbol **N**.

What a force can do

- Make a resting object **move** (start motion).
- **Change the speed** of a moving object (speed up or slow down).
- **Change the direction** of motion.
- **Change the shape** of an object (e.g. pressing a balloon).

A force may cause one or several of these effects at once.

Force is an interaction

When your hand pushes a table, your hand and the table interact — and your hand feels a force too. The moment the interaction stops, the force disappears.

At rest doesn't mean no force: an object can be at rest because the forces on it are **balanced** (cancel out).

Where students lose marks

Trap 1 — A force needs two objects. One object alone can't exert a force on itself; there must be an interaction.

Trap 2 — Change of shape is also an effect of force. Don't forget it — pressing/stretching changes shape.

Trap 3 — Unit is newton (N), small "n". Write "5 N", not "5 Newtons" with a capital letter.

At rest: a still object may have balanced forces acting on it, not zero force.

Model answers — write them like this

Example A. State the effect of force: (i) a fielder stopping a ball (ii) pressing dough flat.

Step 1: Stopping a moving ball → changes its speed (to zero).

Step 2: Pressing dough → changes its shape.

∴ (i) change of speed (ii) change of shape.

Example B. When you hit a moving ball with a bat, what does the force do?

Step 1: The ball was moving one way; the bat pushes it the other way.

∴ The force changes the ball's direction (and speed).

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